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Competitive effects of anion mobility and viscous forces on the interactions of hyaluronic acid and alginic acid with hofmeister salts

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A R T I C L E I N F O

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A B S T R A C T

The specific ion effect takes into account the hydration, mobility, and kosmotropic/chaotropic effects. However, another important aspect to consider is the solution viscosity, which influences ion mobility. A limited number of publications have considered the impact of specific ion effects on polyelectrolytes. However, none of the studies have considered the viscous effect of polyelectrolytes. This study aims to investigate the impact of kosmotropic and chaotropic anions on negatively charged alginic acid and hyaluronic acid using dynamic light scattering, rheology, and molecular dynamic simulations. Different concentration regimes and the change in the chain conformations in these regimes, along with the viscosity scaling and impact of the salts on the polyelectrolytes, have been discussed. An interesting finding regarding the solution viscosity was reported. The solution viscosity leads to the reversal of the diffusivity and viscosity trend of the polymer with kosmotropes and chaotropes. A threshold is also identified above which viscous forces dominate. Specific ion effects dominate below the threshold. It was concluded that the specific ion effect shows a competitive nature with the viscous forces in the case of a high-viscosity polyelectrolyte solution, which becomes an important aspect to consider for unveiling such interactions. Prior knowledge of the conformations of polymers can be used for designing drug delivery/nutrient delivery systems and for understanding phase separation behavior. Considering the viscosity of polymer solution becomes an important factor which, in turn, impacts the anion's mobility and ultimately influences the polymer conformations.

1. Introduction

Alginic acid (AA) and hyaluronic acid (HA) are carbohydrate polymers widely used in various applications. Apart from carbohydrate polymers, they can be categorized as negatively charged polyelectrolyte systems. Polyelectrolytes are charged macromolecules/polymers that have similarly charged monomer groups on their backbones (Muthukumar, 2017). AA and HA have carboxylic acid groups on their monomers, which deprotonate in their aqueous solution with a pH above their respective pKa (3.5 for alginate (Francis et al., 2013) and 3 for HA (Brown & Jones, 2005). This causes the functional groups on the backbone to gain negative charges. Hence, AA and HA are classified as polyelectrolytes, more specifically as polyanions.

The polymers exhibit extended or compact conformations in their respective solution based on the governing forces. Electrostatic forces, dipolar interactions, van der Waals forces, and H-bonds are the primary forces governing these conformations (Basu et al., 2024; Das, Basu, et al., 2024a, Das et al., 2024a,b, Das & Majumdar, 2024a; DOBRYNIN, 2008; Dobrynin et al., 2004; Sayko et al., 2021). However, for charged polymers, such as polyelectrolytes, the electrostatic force plays a dominant role (owing to its relatively higher strength than other forces) in controlling the conformations. AA and HA exhibit an extended conformation in their aqueous solution due to the inter-chain and intra-chain electrostatic repulsions among the negatively charged functional groups of their backbones (Muthukumar, 2017). These electrostatic interactions among the polymer chains can be altered by the addition of salts, which in turn affect their conformations.

The addition of salts to a solution of a charged polymer leads to additional electrostatic interactions, i.e., screening. It is an already established fact that salts screen the electrostatic repulsions between the charged functional groups of the polyelectrolytes, which in turn results in the collapse of the chains (Muthukumar, 2017). However, the type of salt ion has an additional impact on the conformation of the polymers. Salt ions are majorly categorized into two categories: kosmotropes and chaotropes (Filova et al., 2020; Yu et al., 2019). The kosmotropes, being the larger ions, have higher attractive dispersion forces, which result in

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more interactions with the water molecules. Hence, these ions have a higher hydration radius, which results in reduced mobility in the solution.

On the contrary, the chaotropes being the smaller ions have a lower hydration radius and, thus, higher mobility (Parsons et al., 2011). Apart from the mobility and hydration, another important factor to consider for these interactions is the repulsion from peptide bonds for kosmotropes (Balos et al., 2016; Giner et al., 2007). Moreover, the peptide bonds serve as binding sites for chaotropes (Balos et al., 2016; Rogers et al., 2022). These interactions, along with the electrostatic effect of the ion, play a pivotal role in governing the conformation and phase separation of proteins. The repulsions/attachments from/to the peptide bonds of proteins, along with the mobility and hydration radius of anions, contribute to the impact of ions on the protein chains, which is known as the specific ion effect (Mazzini & Craig, 2017; Parsons et al., 2011; Parsons & Ninham, 2011; Paterová et al., 2013; Rembert et al., 2012).

The specific ion effects are a widely studied phenomenon in protein systems (Medda et al., 2012, 2015, Reichheld et al., 2017; Yu et al., 2019). Proteins can also be classified as charged polymer systems. However, proteins are categorized into a different class of charged polymers, i.e., polyampholytes, which have both positively and negatively charged groups on their backbones, along with a peptide linkage between the monomers. AA and HA are polyelectrolytes with glycosidic linkages between monomers. Hence, the specific ion effect on AA and HA would be different when compared to proteins (Tatini et al., 2017).

Few cases of the impact of specific ion effects on polyelectrolytes have been reported in the literature. Musilová et al. (2019) studied the impact of chaotropes and kosmotropes on the chain size and viscosity of hyaluronan solutions. The viscosity and chain size measurements inferred higher expansion of hyaluronan coils in the case of kosmotropes compared to chaotropes. Jacobs et al. (2021)investigated the influence of multivalent cations on negatively charged carboxymethyl cellulose and polystyrene sulfonate solutions. The chaotropes show a higher increase in solution viscosity due to higher cross-linking and entanglements and counter-ion-induced association in the polymer chains than kosmotropes. Luo et al. (2012) showed the impact of salts on the diffusivity of quaternized poly (4-vinyl pyridine). The diffusivity with the chaotropic salt anion showed a higher value than the kosmotropic anions.

Xu et al. (2019) studied the impact of divalent cations on negatively charged polystyrene sulfonate brushes. The chaotropic cation showed a higher and profound impact on the structure of polyelectrolyte brushes. Kou et al. (2015) showed the impact of anions on the positively charged Poly [2-(methacryloyloxy) ethyl trimethylammonium chloride] polyelectrolyte brushes. The chaotropic anions showed stronger ion-pairing interactions with the oppositely charged brush than kosmotropes. At higher concentrations, the chaotropes also showed a collapse of polymer chains in the brush. Salomäki and Kankare (2008) considered the impact of the different anions on the swelling behaviour of poly(diallyl dimethylammonium) and poly(4-styrene sulfonate) multilayers. The chaotropes showed lower swelling of the polyelectrolyte multilayers compared to kosmotropes. Ehtiati et al. (2022) also studied the swelling behaviour of polyelectrolyte brushes in the presence of different anions. Positively charged poly[2-(1-butylimidazolium-3-yl) ethyl methacrylate] polyelectrolyte brushes were used to study the same. The swelling of the brushes increased with the increasing kosmotropic nature of the anions. Another study showed the impact of salts on the separation and recovery of chitosan from its acidic solution. The kosmotropes showed a more potent salting-out behavior of chitosan than chaotropes (LeHoux & Dupuis, 2007).

The pieces of literature mentioned above have mostly considered the impact of different salts on polyelectrolyte brushes. Very few studies have considered the impact of salts on the polyelectrolyte chains in solution. The specific effect of such salt ions should ideally depend on the solution viscosity, as the mobility of such ions is also affected. To the best of our knowledge, no literature has presented the impact of solution viscosity induced by the polymers on the specific ion effect of the anions. This article aims to focus on this uncharted territory. NaCl, $NaNO_3$ (two chaotropes), Na_2SO_4 and CH_3COONa (two kosmotropes) are considered for the investigation of their impact on HA and AA. First, the impact of salts on the diffusivity and viscosity of HA and AA was discussed. All-atom molecular dynamics (MD) simulations were also conducted on a single chain of HA and AA with different salt ions to corroborate the experimental results. The results showed that the higher mobility of the chaotropic anion resulted in a higher electrostatic repulsion with the negatively charged HA chains. The results obtained for HA were further validated by examining the impact of the salts on low-viscosity AA (LVAA). The specific ion effect of the salts on LVAA was the same as for the case of HA, as HA is also a low-viscosity polymer. MD simulations of single-chain AA and different salts again authenticated the results. Finally, the impact of these salts on high-viscosity AA (HVAA) was investigated. It was concluded that the viscosity of the polymer solution impacts the specific ion effects. The trend in diffusivity and viscosity of the polymer solutions with the addition of salts reverses with an increase in solution viscosity beyond a threshold point. This reversal beyond a threshold was further confirmed by examining the diffusivity and viscosity of LVAA at higher concentrations. The same could not be proved using the MD simulations, as the introduction of viscous forces in the simulation is a difficult proposition for GROMACS (simulation software). Hence, the single-chain MD simulations were conducted to investigate and corroborate the experimental results at low polymer concentrations and where viscous forces were absent. Hence, it is concluded that the specific ion effect induced by the salt ions shows a competitive effect with the solution viscosity induced by the polymer. The specific ion effect dominates in the low concentration and viscosity regimes, whereas viscous forces dominate in the high concentration and viscosity regimes beyond a threshold.

2. Experimental section

2.1. Materials

NaCl, $NaNO_3$, Na_2SO_4, CH_3COONa, and hyaluronic acid were purchased from SRL Chemicals (India). High-viscosity and low-viscosity alginic acid sodium salt was procured from Alfa Aesar (USA). The viscosity of the high-viscosity 0.5% (w/v) alginate solution at pH 7.5 is 167 $\pm$ 22 cP at 30 °C and 100 s^{-1} shear rate. It has a molecular weight of 310 $\pm$ 50 kDa and a zeta potential value of -30 ± 3 mV. The viscosity of 0.5% (w/v) low viscosity alginate solution at pH 7.5 is 3.55 $\pm$ 0.5 cP at 30 °C and 100 s^{-1} shear rate. It has a molecular weight of 160 $\pm$ 40 kDa and a zeta potential value of -22 ± 2.1 mV. The FTIR study for alginate was conducted by Bruker Tensor 37 using a KBr pellet. The spectral range was collected between 600 and 4000 cm^{-1} with a spectral resolution of 4 cm^{-1}. 256 scans were performed to ensure the reproducibility of the data. The M/G ratio (Absorbance at 1030 cm^{-1}/Absorbance at 1080 cm^{-1}) was calculated by the method reported in the literature (Beratto et al., 2017; Pragya et al., 2021). The M/G ratio for high-viscosity alginate is 1.68 $\pm$ 0.12, and for low-viscosity alginate is 0.70 $\pm$ 0.15. Hyaluronic acid sodium salt was procured from SRL chemicals. The viscosity of 0.5% (w/v) hyaluronic acid solution at pH 7.5 is 3.167 $\pm$ 0.34 cP at 30 °C and 100 s^{-1} shear rate. The molecular weight of hyaluronic acid is 90 $\pm$ 10 kDa and a zeta potential value of -14 ± 2.4 mV.

2.2. Sample preparation

HA and AA were dissolved in a pH 7.5 solution (prepared by adding NaOH to DI water) at 30 °C. The solution was continuously stirred for 30 min. Different salts (0.1 M) were added to the solution and continuously stirred for another 45 min prior to further analysis.

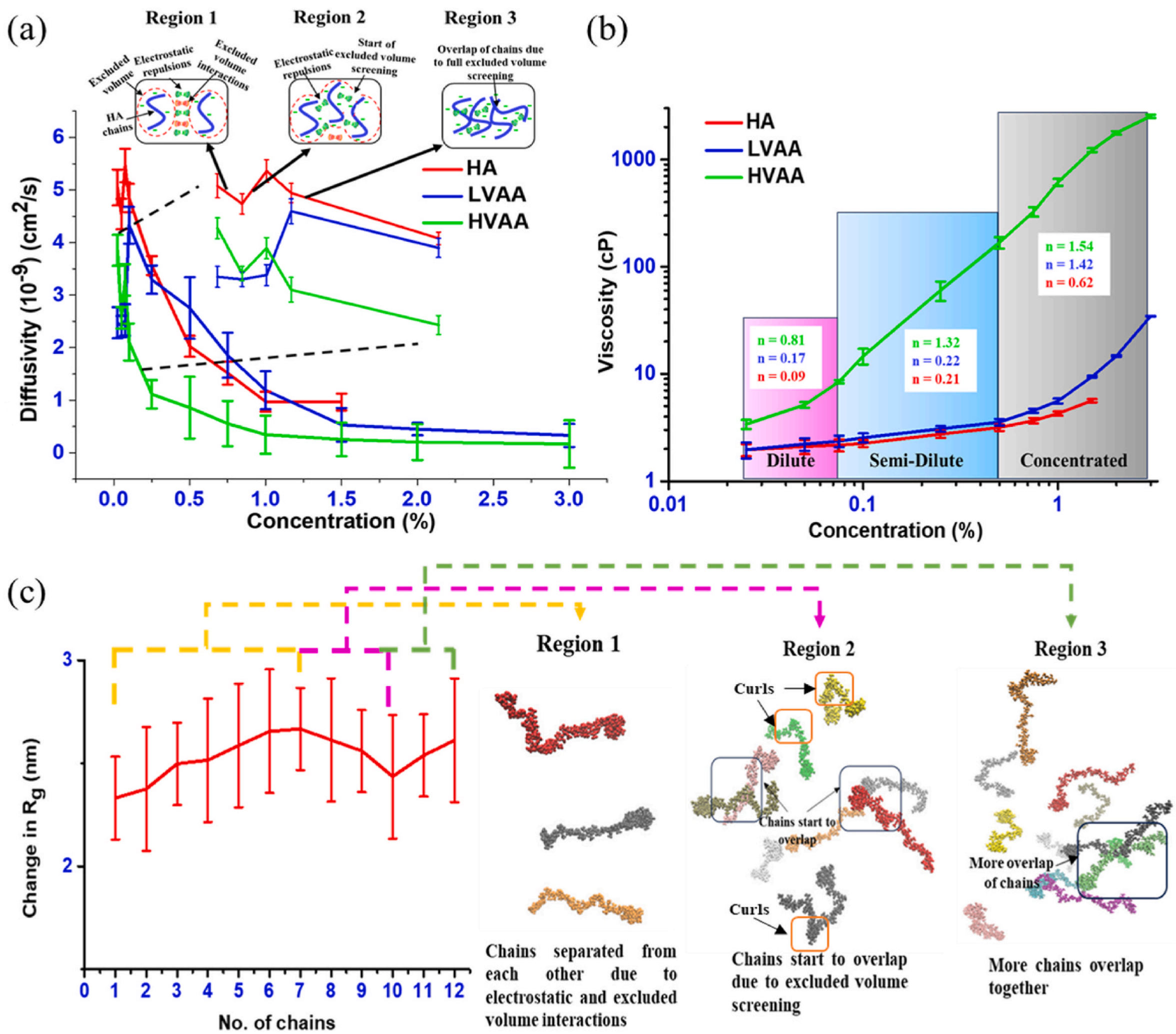

Fig. 1. Changes in chain interactions with an increase in concentration. (a) Changes in diffusivities of HA, LVAA and HVAA with an increase in concentration along with a rough schematic of interactions, (b) Changes in viscosities of HA, LVAA, and HVAA with an increase in concentration, and (c) Changes in R_g of HA with an increase in concentration from MD simulations along with simulation snapshots.

2.3. Dynamic light scattering (DLS)

The physical interactions due to the electrostatic effect of salts were analyzed using the change in diffusivities of the polymer solution with the addition of different salts. The diffusivities were quantified using dynamic light scattering (DLS) (Nano SAQLA, Otsuka Electronics, Japan). The polymer solutions are categorized as colloidal solutions, which exhibit Brownian motion. The scattered light (in DLS) from a suspension of Brownian motion shows intensity fluctuations. These fluctuations are related to the diffusivity of the Brownian particles. These time-scale fluctuations of Brownian motion are related by the auto-correlation function, which is as follows (Ross Hallett, 1994)

$$G_\tau^{(2)} = \langle I(t).I(t+\tau) \rangle \tag{1}$$

where, I(t) is the scattered light intensity and τ is the delay between two time points.

The normalized light intensity ACF $\left(g_\tau^{(2)}\right)$ and the normalized electric field ACF $\left(g_\tau^{(1)}\right)$ are related by Siegert relation as

$$g_\tau^{(2)} = 1 + \beta \left|g_\tau^{(1)}\right|^2 \tag{2}$$

where, β is the coherence factor.

Now, the electric field ACF is expressed as

$$g_\tau^{(1)} = \exp(-\Gamma \tau) \tag{3}$$

Where, Γ is the decay constant and,

$$\Gamma = D_T q^2 \tag{4}$$

where D_T is the translational diffusion coefficient and q is the scattering vector.

Now,

$$g_\tau^{(2)} = 1 + \beta e^{-2\Gamma\tau} \tag{5}$$

which can be substituted as

$$g_\tau^{(2)} = 1 + \beta e^{-2D_T q^2 \tau} \tag{6}$$

which is the correlation function and connects the particle motion with the measured fluctuations.

The studies were conducted at 30 °C for all polymer concentration regimes. Modern-day DLS can measure the changes in diffusivities of polymer samples in dilute and semi-dilute regimes of polymer. All the measurements were performed in triplicates. The scattering angle was 165°, and the laser wavelength was 658 nm. CONTIN algorithm was used for data processing. The backscattering angle was used for the scattering techniques as it reduces the interference of multiple scattering lights from other chains of polymer.

2.4. Viscosity measurements

The viscosity measurements of the polymers were carried out on a Rheometer (Anton Paar, MCR 92). The viscosity was measured at a temperature of 30 °C and a shear rate of 100/s. The cone and plate geometry (25 mm diameter, truncation distance: 0.1 mm) was used for the measurements. Practically, for polymers (non-Newtonian (pseudo-plastic fluids)), whose properties change with concentration, chain size, and pH, the apparent viscosity (which is measured in rheology experiments) is the ideal parameter that can provide an idea of viscosity.

2.5. MD simulation Methodology

The GPU version of Gromacs 2020.4 was used to perform the molecular dynamic simulations (Abraham et al., 2015). OPLS-AA forcefield with three-dimensional periodic boundary conditions (Jorgensen & Tirado-Rives, 1988) was used. The energy minimisation was performed using the steepest descent algorithm ($F_{max} < 1000$ kJ/mol/nm). A 100 ps NPT (constant-temperature, constant-pressure ensemble) simulation was performed with the energy-minimized structure, followed by a 5 ns MD production run with a time step of 1 fs. The equation of motion was integrated using the leap-frog integrator. V-rescale thermostat was used for temperature coupling at 303 K (Bussi et al., 2007). An isotropic Berendsen barostat with a target pressure of 1 bar, coupling parameter of 2 ps and isothermal compressibility of 4.5×10^5 bar^{-1} was used (Berendsen et al., 1984). The production runs were performed using the same thermostat and an isotropic Parrinello–Rahman barostat (same conditions as above) (Parrinello & Rahman, 1981). The simulation conditions were kept similar to the conditions maintained by Kopplin et al. (Kopplin et al., 2021). The water molecules were treated with the TIP4P water model (Mahoney & Jorgensen, 2000). The particle mesh Ewald (PME) method was employed to compute long-range electrostatic interactions (Darden et al., 1993). The cut-off distance for the short-range electrostatic and van der Waals interactions was taken as 1 nm. No bonds were constrained by using the LINCS algorithm. The last 1000 time frames separated by 5 ps of the simulation trajectory from the MD production run were utilised to calculate the average over all the statistics. OPLS-AA forcefield was used for the simulations (Jorgensen & Tirado-Rives, 1988). Other parameters like partial charges on each atom, sigma, epsilon, angle, dihedral and improper coefficients were taken from the original OPLS-AA forcefield parameter files present in the GROMACS directory. The initial molecular configurations were prepared using Avogadro (version 1.2.0) software using SMILES (Simplified Molecular Input Line Entry System) text as input. The molecular topology was created using TopolGen (version 1.1). The atom parameters were modified using the OPLS-AA-defined parameters. VMD (Visual Molecular Dynamics, version 1.9.3) was used for visualising the molecular structure (Humphrey et al., 1996). Single alginic acid and hyaluronic acid chains with 30 monomers were used. A box length of 15 nm was taken for the simulations. An equivalent amount of 0.1 M of salts was added. A 50 ns simulation run was conducted to equilibrate the system with salts.

3. Results

The diffusivity of the polymers was considered to comprehend the impact of specific ion effects and electrostatic interactions induced by the addition of salts. The primary objective of this study was to analyze the specific anion effects of salts on the bulk solution properties of negatively charged polyelectrolytes. The mobility of the salt anions will ultimately impact the electrostatic interactions within the polymer chains. Hence, the changes in the diffusivity of the polymers with the addition of salts to their bulk solution were measured to interpret the ion impact. The hydrodynamic radius (R_H) and/or diffusivity of the polymer are inversely proportional to each other, as known from Stoke-Einstein's equation, i.e., $D = \frac{k_B T}{6\pi\eta R_H}$ where, D is the diffusivity, k_B is the Boltzmann constant, T is temperature, η is the viscosity and R_H is the hydrodynamic radius. Hence, the diffusivity gives a fairly good idea about the polymer size in its bulk solution and their changes with the addition of different salts. This can be further correlated with the changes in R_g, which were determined from MD simulations.

The changes in diffusivity with the concentration of polymers show an initial decrease, followed by a slight increase and then further decrease with concentration (Fig. 1 (a)). The diffusivity shows an initial decrease (increase in R_H) in a very low concentration regime (Region 1 in Fig. 1 (a)). This was due to the presence of electrostatic repulsion and excluded volume interactions among the chains. MD simulations were carried out using HA by adding one chain at each step and calculating the R_g to interpret the impact of concentration. The same trend (as observed for R_H) can be inferred from the increase in R_g from the MD simulations at low concentrations (Region 1 in Fig. 1 (c)). The simulation snapshots also show the chains being far apart from each other. A further increase in concentration (Region 2 in Fig. 1 (a)) leads to the beginning of the screening of excluded volume interactions. However, electrostatic repulsion still played a prominent role. The increase in concentration leads to the unavailability of the free solution space for the polymer chains. Ideally, the addition of more negatively charged chains in infinite space will lead to the chains remaining separated from each other and in an expanded state owing to inter-chain repulsions. However, in the finite space, the inter-chain electrostatic repulsions may lead to curling in the undissociated parts or the parts that do not have the carboxylic acid group for dissociation. Hence, a small increase in diffusivity (decrease in R_H) is observed (due to curling in chains), which is further corroborated by the decrease in R_g from the MD simulations (Fig. 1 (c)). The simulation snapshots also show the overlap of a few chains and other chains being far apart from each other, along with a few curlings in the chains. A further increase in the concentration leads to a change from a dilute to a semi-dilute concentration regime of HA, where the chains overlap with each other, and the excluded volume interactions are fully screened.

Hence, due to the overlapping of chains, a decrease in diffusivity (increase in R_H and R_g) (Region 3 in Fig. 1 (a) and (c)). The simulation snapshots also show a higher overlap of the chains. This change from a dilute to a semi-dilute regime can also be inferred from viscosity scaling (Fig. 1 (b)). Hence, the initial increase followed by a decrease in chain size occurs in the dilute regime due to screening of excluded volume effects. This is followed by an increase in chain size due to the overlapping of chains in the semi-dilute and concentrated regimes.

Another important factor that affects interactions in polyelectrolytes is counter-ion condensation, which is a well-known phenomenon. The counter-ions present in the solutions get attracted to the polyelectrolyte backbone. Higher counter-ions can even lead to the reversal of the surface charge of the polyelectrolytes. However, for the above-discussed case, a higher number of counter ions are not present to induce the charge reversal of HA or AA as the solution pH considered for our case is 7.5 (near the physiological pH). To confirm the absence of sufficient counterions to induce charge reversal, the zeta potential measurement for AA was conducted at a 0.4% (w/v) concentration (semi-dilute). The zeta potential value was -20 ± 2.5 mV. Hence, the negative zeta potential confirmed the presence of negatively charged groups on the alginate chains.

Now, at low concentrations of HA and AA, again, the possibility of charge reversal due to counter-ion condensation arises as fewer alginate chains are present, and the counter-ion present may be enough to induce the charge reversal. However, this also has a limited possibility, as in dilute solution cases, the entropic penalty related to counter-ion condensation is very high. Hence, free counter-ions dominate in dilute polymer concentration regimes (Smiatek, 2020). The same was confirmed again with zeta potential measurements. The zeta potential value of 0.05% (w/v) AA concentration (dilute) was −12 ± 3.4. mV.

3.1. Interaction of HA with salts

The addition of salts to negatively charged polyelectrolytes leads to the screening of electrostatic repulsion. The Na^+ salt cations will result in the screening of the negatively charged groups of HA. However, the anions ideally should also have an impact on the chain size. The anions can also diffuse inside the polymer chains through the sites of screened dissociated groups or undissociated groups, as there were no repulsions present. These anions can provide additional electrostatic repulsion upon interaction with the negatively charged unscreened functional

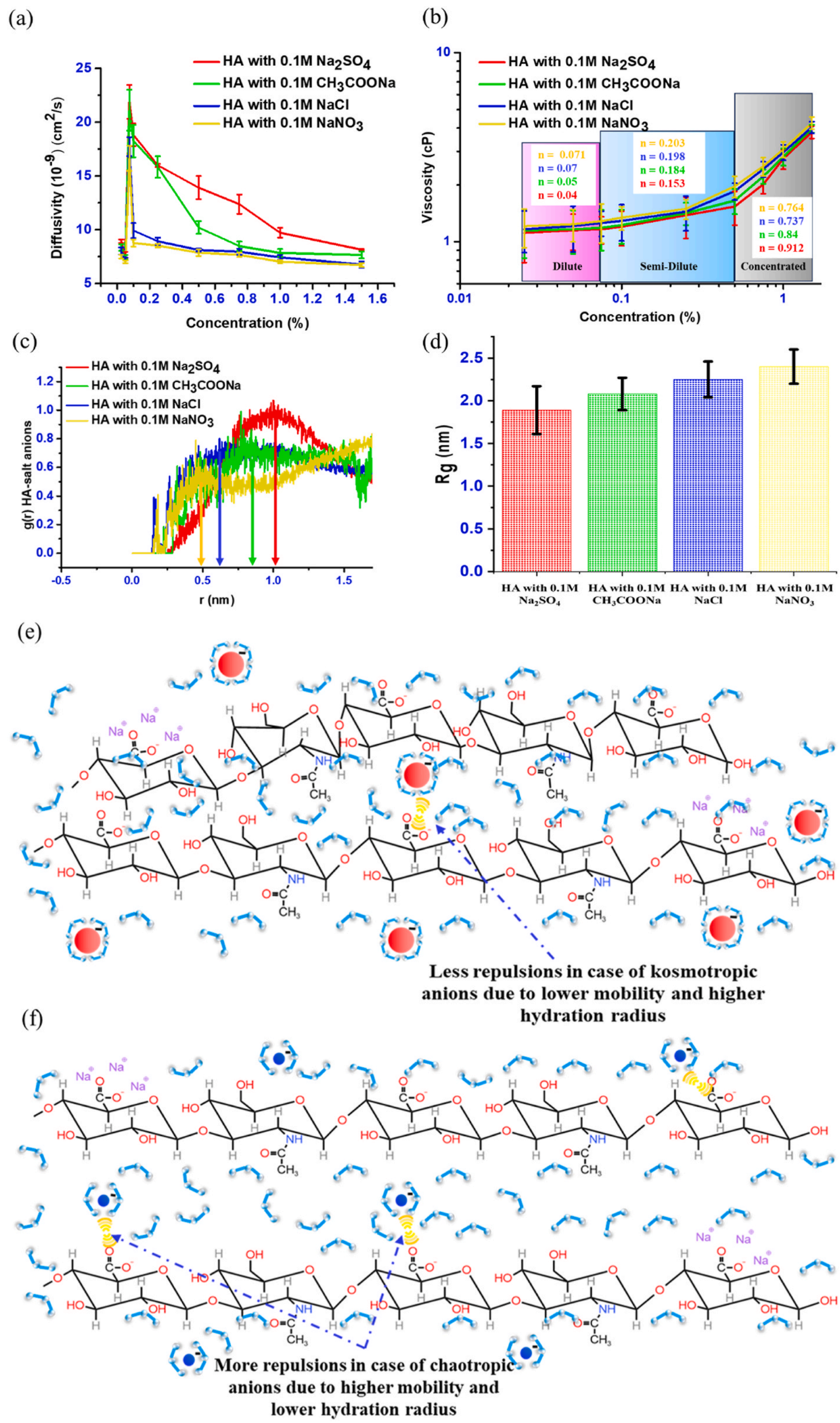

Fig. 2. Interactions of HA with salts. (a) Changes in diffusivities of HA with 0.1 M salts, (b) Changes in viscosities of HA with 0.1 M salts and scaling in different regimes, (c) RDF of HA and salt anions from simulations, (d) Changes in R_g of HA with 0.1 M salts from simulations, (e) A rough schematic of interactions between HA and kosmotropes, and (f) A rough schematic of interactions between HA and chaotropes (more Na^+ around certain COO^- represent screening of charges).

groups of HA. However, the change in the chain size due to these repulsions is dependent on the specific ion effect of the anion. Now, for the case of Na_2SO_4, the charge density of the solution would be different due to the presence of extra Na^+ and the bivalency of the SO_4^{2-}. 0.1M Na_2SO_4 was considered on the premise that an equal number of anions of all salts were introduced into the polymer solution. Now, the double Na^+ in Na_2SO_4 can induce higher screening of the negatively charged polyelectrolytes. However, the bi-valency of SO_4^{2-} induces more repulsion from the unscreened negatively charged groups of the polyelectrolytes.

Fig. 2 shows the impact of the salt anions on HA. The kosmotropes (SO_4^{2-} and CH_3COO^-) have a higher hydration radius and lower mobility, and the chaotropes (Cl^- and NO_3^-) have a lower hydration radius and higher mobility (Parsons et al., 2011). Hence, the chaotropes can freely diffuse through the polymer chains than the kosmotropes, which results in more repulsions in the case of chaotropes. Thus, the chain size of HA with chaotropes is larger than that of HA with kosmotropes. This can be inferred from the trend of the diffusivities of HA with the addition of 0.1 M salts. HA with 0.1 M chaotropic salts has a higher R_H (lower diffusivity values) than HA with 0.1 M kosmotropic salts (Fig. 2 (a)). This trend continued over the entire concentration regime of HA. The MD simulations of HA with salts were also carried out to comprehend the impact of the salts on HA. The values of R_g calculated from the simulations (Fig. 2 (d)) show the same trend with $R_{H,}$ which further upholds the experimental results. To further substantiate the claim about the interactions between salt anions and HA, the RDF (Radial distribution function) analysis between HA and salt anions was performed from the MD simulations (Fig. 2 (c)). RDF gives an idea about the number of target atoms present in the vicinity of the reference atoms. It can be inferred that the chaotropic anions are higher in the vicinity of HA chains than kosmotropic anions, implying higher interactions between chaotropic anions and HA chains. Hence, it can be concluded that the mobility of the anions and the electrostatic repulsion from the anions affect the conformations of HA in the entire concentration regime (Schematics shown in Fig. 2(e) and (f)). The viscosity scaling and the scaling exponents in the three regimes of HA are shown in Fig. 2 (b). HA with kosmotropes exhibited lower viscosity values than HA with chaotropes. As discussed above, chaotropes can diffuse inside the HA chains and induce more electrostatic repulsion, resulting in higher R_H or R_g. This resulted in a higher overlap of chains, which ultimately increased the viscosity of the polymer solution.

The single factor ANOVA analysis with 95 % confidence suggests that there is a significant difference (p-value = 0.00168 < 0.05 and F value = 15.114 > F critical = 4.066) in diffusivities of HA with salts in the dilute regime (0.025 % (w/v)) of HA. The same was true for the diffusivities of HA with salts in the semi-dilute regime (0.5 % (w/v)) of HA (p-value = 0.0000156 < 0.05, F value = 92.52 > F critical = 4.066).

3.2. Interactions of LVAA with salts

Fig. 3 (a) shows the changes in the diffusivity of LVAA with different

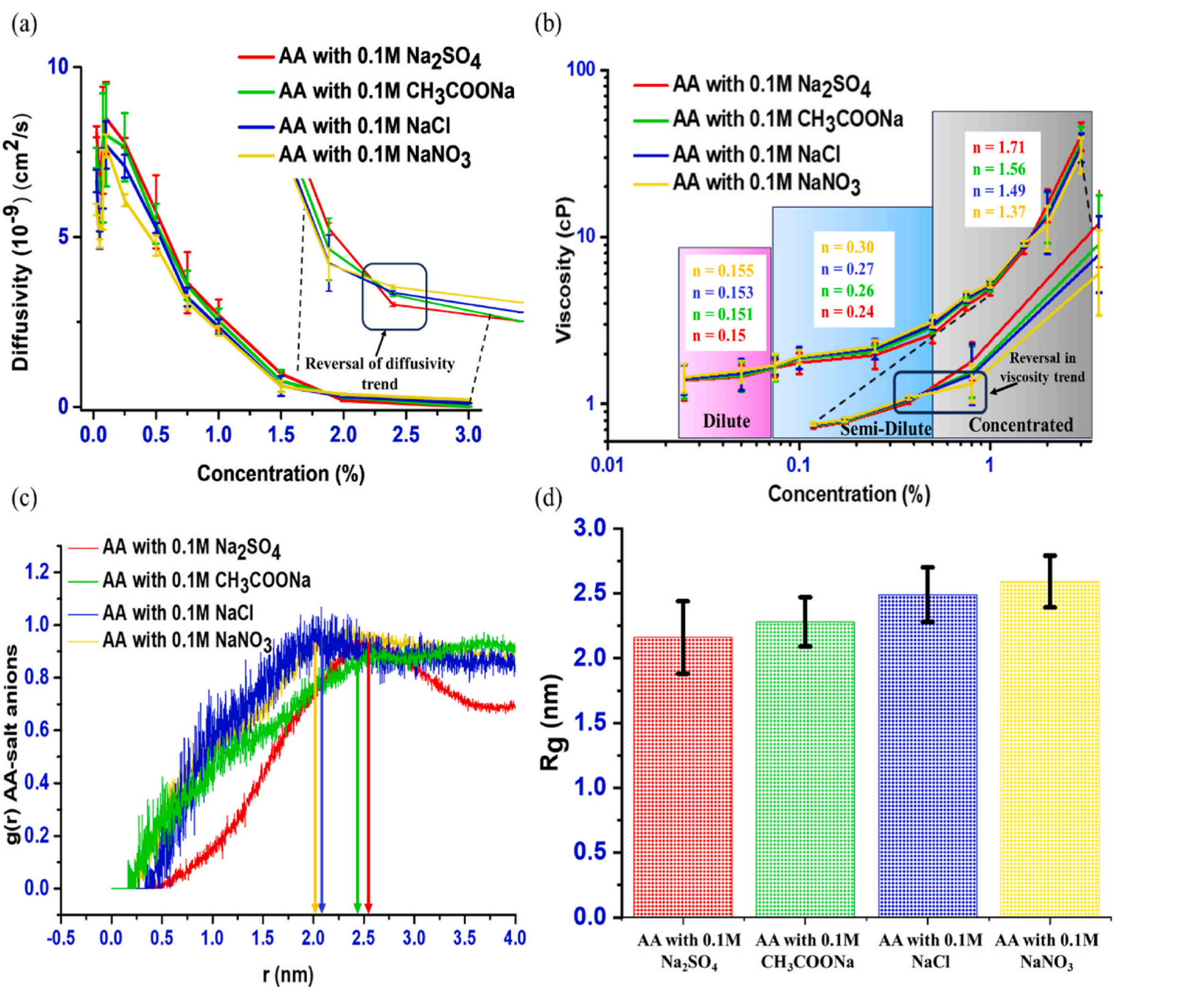

Fig. 3. Interactions of LVAA with salts. (a) Changes in diffusivities of LVAA with 0.1 M salts, (b) Changes in viscosities of LVAA with 0.1 M salts and scaling in different regimes, (c) RDF of AA and salt anions from simulations, (d) Changes in R_g of AA with 0.1 M salts from simulations.

salts. Similar to the case of HA, the interactions between the kosmotropic salts and chaotropic salts with LVAA were the same. The same can also be concluded from the RDF plots of AA and the anions and the changes in R_g from the MD simulations.

Similar to the MD simulations of HA, the AA chain showed the same trend as the salt anions. The kosmotropes having higher hydration radii and lower mobilities are unable to diffuse inside the chains. Hence, fewer electrostatic repulsions occur between the kosmotropic anions and unscreened charged groups of LVAA, thus justifying a trend similar to that of HA. Fig. 3 (b) shows the scaling of viscosity with LVAA and different salts.

The single factor ANOVA analysis with 95 % confidence suggests that there is a significant difference (p-value = 0.000000297 < 0.05 and F value = 140.18 > F critical = 4.066) in diffusivities of LVAA with salts in the dilute regime (0.025 % (w/v)) of LVAA. The same is true for diffusivities of LVAA with salts in the semi-dilute regime (0.5 % (w/v)) of LVAA (p-value = 0.000127 < 0.05 and F value = 28.54 > F critical = 4.066) and concentrated regime (2% (w/v)) of LVAA (p-value = 0.000000106 < 0.05 and F value = 182.19 > F critical = 4.066). A reversal in the interactions was observed in the concentrated regime. Hence, the ANOVA analysis was also conducted to confirm the statistical significance of the results.

A time-series DLS was conducted at 1% (w/v) concentration and 2.5% (w/v) concentration of LVAA to study the impact of the mobility of anions on the electrostatic repulsion. Since these repulsions are dynamic, no attractive force is present to bind the anions at a specific site. Thus, the anions may again diffuse out of the polymer chains at low concentrations due to their mobilities. However, these electrostatic forces are long-range forces, and thus, these forces are continuously acting on the negatively charged polymers and the anions are continuously diffusing in and out of the polymer chains in its non-equilibrated state. Hence, to minimize the free energy, the polymer chains ultimately expand. Fig. 4 shows the changes in diffusivity of LVAA with time with the addition of salts. The samples were initially stirred for 5 min for the dissolution of salts. DLS was performed every 1.5 min to check for changes in diffusivities until the samples reached equilibrium (no change in diffusivity values). It is evident from the figures that the fluctuation in the diffusivities is much higher for the 1% (w/v) concentration of LVAA.

Moreover, at a 1% (w/v) concentration, the equilibration time was also higher for chaotropes than for kosmotropes. As explained above, a higher mobility of chaotropes leads to a more dynamic behavior and higher electrostatic repulsions with the negatively charged LVAA. Hence, more time is required for equilibration of the polymer with chaotropes and a higher R_H owing to more repulsions. On the contrary, the equilibration time is almost similar for all the cases at 2.5% (w/v) concentrations. The impact of anions on polymers at higher concentrations (2.5% (w/v)) is explained below. A peculiar observation can also be inferred from Fig. 3(a) and (b). A reversal in the trend of salt interactions with LVAA beyond a polymer concentration of 1.5% (w/v) was observed. The same can be inferred from the viscosity studies. To further substantiate this observation, salt interactions with HVAA were studied. The study (in Fig. 4) was conducted to check the equilibration dynamics of the polymer. The initial minutes, where there are higher fluctuations in the diffusivity values, represent the unequilibrated state of the polymer, and the final values are at equilibrium, which are

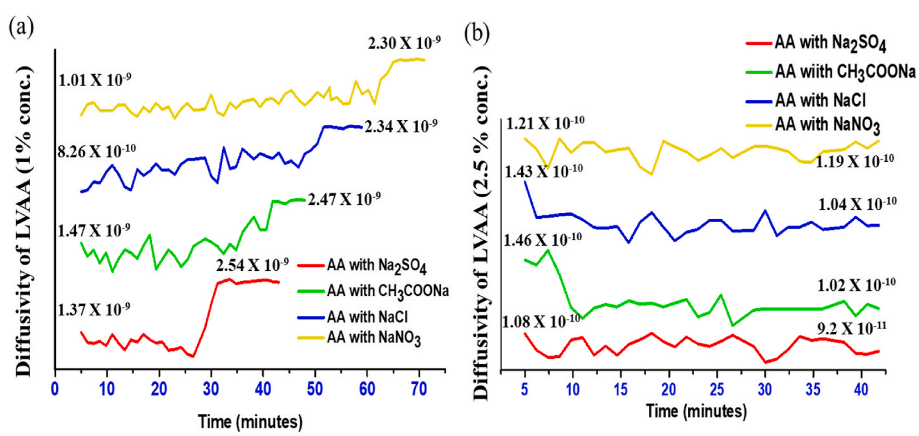

Fig. 4. Time-Series DLS of LVAA with salts. (a) Changes in diffusivities of LVAA (1% (w/v) concentration) and 0.1 M salts with time, (b) Changes in diffusivities of LVAA (2.5% (w/v) concentration) and 0.1 M salts with time.

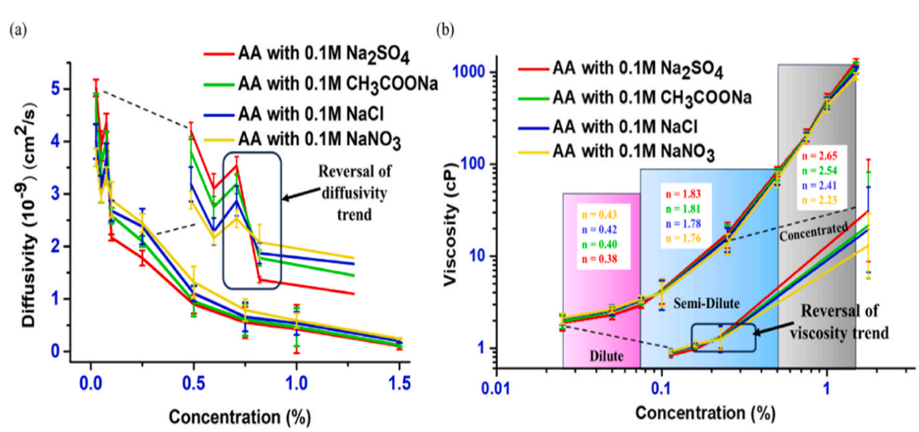

Fig. 5. Interactions of HVAA with salts. (a) Changes in diffusivities of HVAA with 0.1 M salts, (b) Changes in viscosities of HVAA with 0.1 M salts and scaling in different regimes.

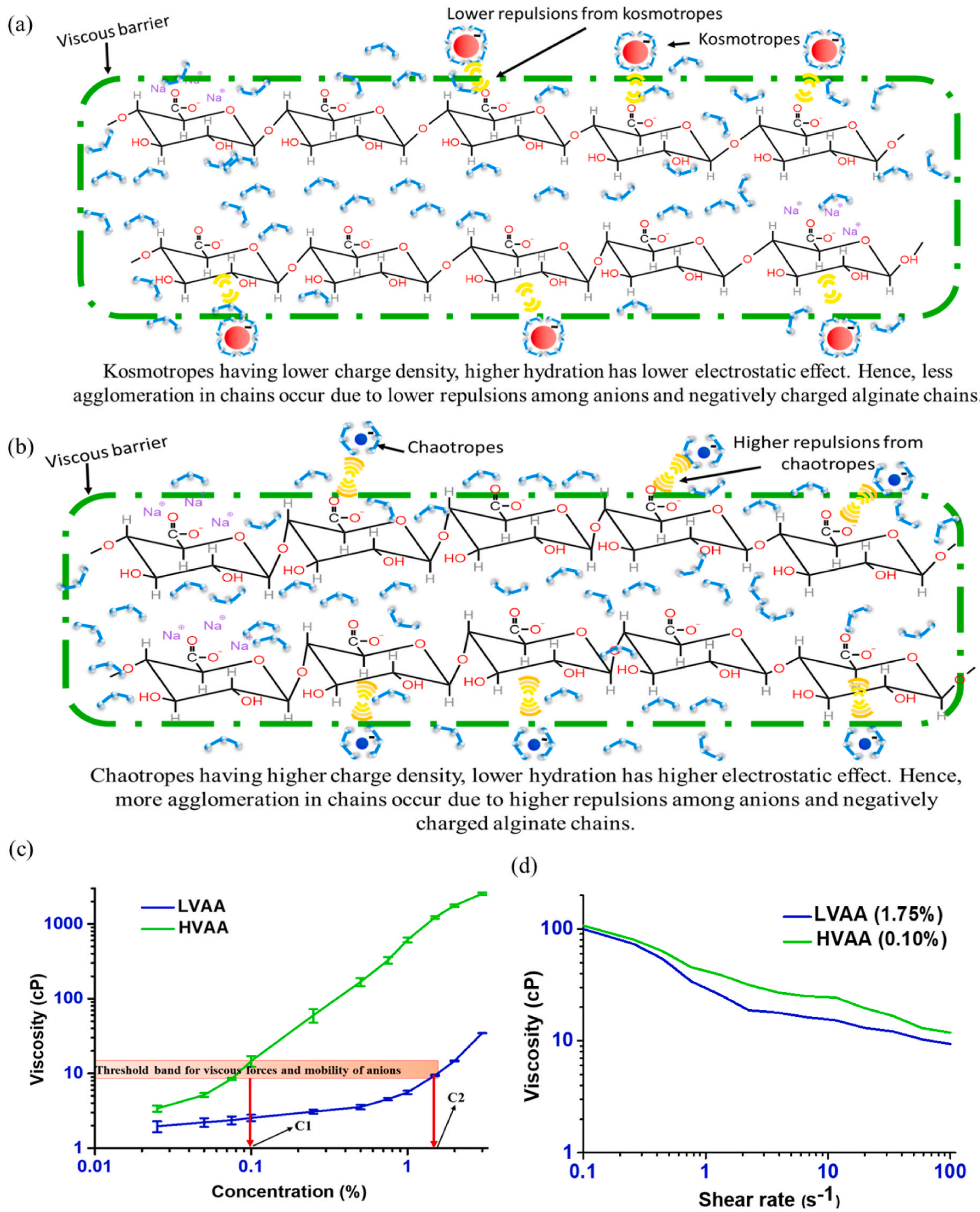

Fig. 6. Impact of viscosity on the salt interactions and AA. (a) A rough schematic of interactions between kosmotropes and AA in high viscosity region, (b) A rough schematic of interactions between chaotropes and AA in high viscosity region (more Na^+ around certain COO^- represent screening of charges), (c) An identification of the threshold region of viscous forces and mobility of anions (C1: Concentration of HVAA at which reversal of salt interactions occurs and C2: Concentration of LVAA at which reversal of salt interactions occurs), and (d) viscosity of LVAA and HVAA at different shear rates.

considered in the data plotted in Fig. 3.

3.3. Interactions of HVAA with salts

Fig. 5(a) shows the changes in the diffusivity of HVAA with different salts. Similar to the case of HA and LVAA, the interactions between the kosmotropic salts and chaotropic salts with HVAA were the same in the dilute regime. Fig. 5 (b) shows the scaling of viscosity with HVAA and different salts.

The reversal of salt interactions was also observed in the case of HVAA. The viscosity of the polymer solution must be considered to explain this phenomenon. At higher polymer concentrations, the solvent (water) is bound to the polymer chains. The hydrodynamic radius of the polymers needs to be satisfied with the solvent in its solution. Hence, the polymer induces the viscous effect and thus ultimately increases the viscosity of the solution. Since the ions are all interacting in the bulk

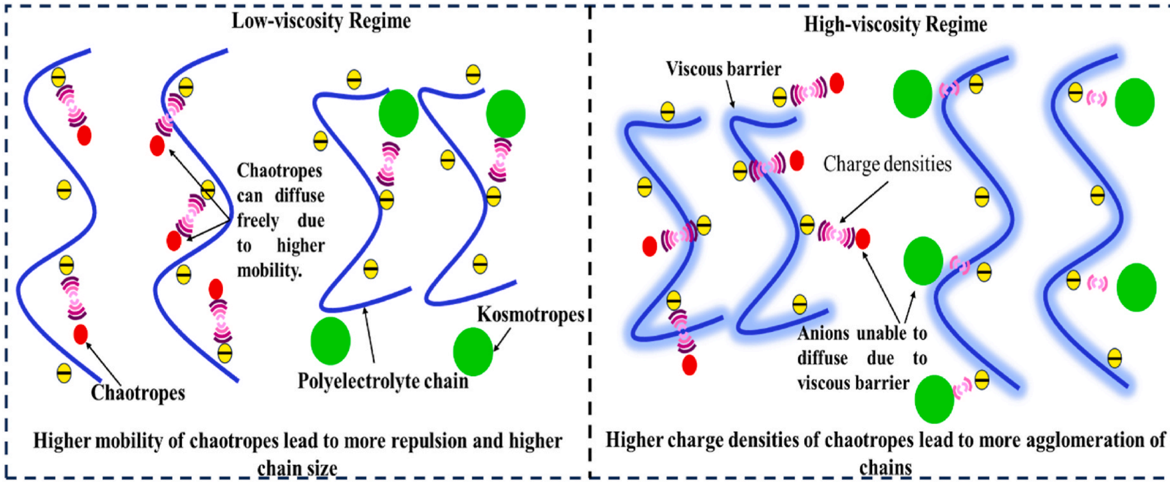

Fig. 7. A rough schematic of the interactions at low viscosity and high viscosity regime of polyelectrolytes.

solution, their mobility will be affected by the viscosity induced by the polymer solution. The high concentrations of LVAA and semi-dilute concentrations of HVAA have high solution viscosity. This induces a viscous barrier for the ions, and hence, the mobility of both kosmotropic and chaotropic anions is affected. Now, both types of anions are unable to diffuse inside the polymer chains, and the ions will interact from the external vicinity of the polymer chains. However, the electrostatic interactions, being the dominant, stronger and long-range force, can interact and change the polymer conformation. The kosmotropes have lower charge densities, and the chaotropes have higher charge densities. Also, the hydrations of chaotropic anions are less compared to kosmotropes. Hence, the electrostatic repulsion of chaotropes with negatively charged polymer groups is stronger than that of kosmotropes because the higher charge density and lower hydration radius of chaotropes lead to higher electrostatic forces. Thus, the chaotropes tend to agglomerate the chains more compared to the kosmotropes. This explains the reversal in the diffusivity and viscosity trends in the high-viscosity regime. As chaotropes have a higher tendency to force the chains together when compared to kosmotropes from the external vicinity, higher diffusivity (lower R_H) is reported for the LVAA and HVAA with chaotropes in a high viscosity regime. A rough schematic is shown in Fig. 6(a) and (b). Since the mobility of the anions is less in the viscous regime, the time required for equilibration of the polymer chains with the salts was less, and much less fluctuations in diffusivity were also observed (Fig. 4(b)).

The single factor ANOVA analysis with 95 % confidence suggests that there is a significant difference (p-value = 0.000241 < 0.05 and F value = 23.84 > F critical = 4.066) in diffusivities of HVAA with salts in the dilute regime (0.025 % (w/v)) of HVAA. The same is true for diffusivities of HVAA with salts in the semi-dilute regime (0.5 % (w/v)) of HVAA (p-value = 0.00099 < 0.05 and F value = 15.87 > F critical = 4.066). Now, this viscous force must have a threshold beyond which it starts to dominate. Below this threshold, the mobility of the anions dominates, and the anions are free to move. Fig. 6 (c) shows the threshold band of viscosity for the mobility and viscous barrier induced by the polymer.

From the figure, it can be inferred that a polymer solution viscosity of 8–11 cP (@ 100 s^{-1} shear rate) serves as a threshold for the viscous barrier and mobility of anions. For LVAA, a concentration of 1.75% (w/v) served as the point of reversal for salt interactions, and the same was observed at a concentration of 0.1% (w/v) for HVAA. The viscosity of both polymers was 8–11 cP (@ at s^{-1} shear rate of 100 s-1), which served as the threshold band. Hence, it can be concluded that the viscous forces and specific ion effects exhibit a competitive nature, with each dominating at the respective regimes of the threshold. Fig. 6 (d) shows the viscosity of LVAA and HVAA at different shear rates. Hence, the plot shows the threshold of viscosity and mobility at different shear rates for LVAA and HVAA. Fig. 7 depicts summary of the above discussed interactions.

The above-discussed results and interactions between negatively charged polyelectrolytes and salts hold for polyelectrolytes in solution without surface charge reversed state due to counter-ion condensation. The investigation of such interactions between polymers and salts can be used to design and develop drug delivery systems (Brown & Jones, 2005; Das et al., 2022, 2023; Das, Giri, et al., 2024; Das & Majumdar, 2024b; Mondal et al., 2020; Panigrahi et al., 2024; Priya James et al., 2014) which can be further extended to nutrient/protein delivery (Lin et al., 2024; Wang et al., 2024; Wu et al., 2024). Another aspect of such interactions is the phase separation behavior. The inclusion of higher concentrations of salts leads to liquid-liquid phase separation and, finally, to solid-liquid phase separation (Lee & Muthukumar, 2009). Thus, understanding the dynamics of polyelectrolytes with these salts can lead to unlocking the potential of such interactions in their phase behaviors, which can further be correlated with intercellular phase transitions (DNA and/or RNA) and disease models (Guo et al., 2021; Nordenskiöld et al., 2024).

4. Conclusions

The salt interactions with negatively charged polyelectrolytes are dependent on the mobility of salt ions and the hydration radius (specific ion effects). Along with these, the viscosity of the polymer solution also has an impact. The viscous barrier induced by the polymer results in the reversal of the interactions between the anions and negatively charged groups of the polymer. This viscous barrier arises from a viscosity range of 8–11 cP (@ 100 s^{-1} shear rate) for polymer solutions. The specific ion effects dominate below this threshold of viscosity, and the anions are free to interact with the chains and induce repulsions from within the polymer chains. In this region, the chains with kosmotropes are lower in size than chaotropes as more chaotropes can diffuse inside the chains and result in higher electrostatic repulsions. Beyond the viscous threshold, the viscous forces dominate. The anions become relatively immobilised due to the high solution viscosity. The charge density and the hydration radius of the anions impact the interactions. The higher charge density and hydration of chaotropes lead to higher electrostatic forces. Thus, chaotropes tend to agglomerate the polymer chains more than kosmotropes in this region. These types of interactions hold for anions and negatively charged viscous polymer solutions, where the polymers are not in their charge-reversed state owing to counter-ion condensation. Hence, it can be concluded that viscosity should also be considered to unveil the impact of specific ion effects on charged polyelectrolytes, which can affect the dynamic equilibrium of the polymer solutions, as well as other processes that fall in the Hofmeister paradigm.

CRediT authorship contribution statement

Sougat Das: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Saptarshi Majumdar:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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